ples), and derive answers from provided documents. XQA (Liu et al., 2019a), one of the few retrieval-independent QA datasets, offers cloze-style questions, leveraging Wikipedia's daily questions and entity answers to populate document-independent answers. TyDi (Clark et al., 2020), like MKQA, focuses on typological diversity in its wide language selection. While TyDi offers a more natural distribution of questions, its annotations are based on the retrieval system used by the authors (Google search); hence their answers are actually start and end indices for spans of text within a given passage. Xor-QA (Asai et al., 2021) explores cross-lingual subtasks by re-annotating 40k TyDi examples, over 7 languages, sourcing answers from English documents and translating them back to the target language. Many of these multilingual resources have been bundled into cross-lingual benchmarks, such as XTREME (Hu et al., 2020) and XGLUE (Liang et al., 2020).

2.1 Comparison to Native Speaker Datasets

There are key advantages to datasets such as TyDi (Clark et al., 2020) and Xor-QA (Asai et al., 2021) which use native speakers questions, particularly in the naturalness and cultural authenticity of the corpora. However, there are also key disadvantages to these datasets which MKQA circumvents with language alignment, to provide more challenging and fair model evaluations across languages.

TyDi (Clark et al., 2020) and MKQA both target high typological diversity, highlight the importance of sourcing realistic questions (with answers unseen), and incorporate a broader distribution of question types than competing datasets (including "No Answer" and "Yes"/"No" answers). There are three main differences between MKQA and TyDi: (a) question alignment across languages, (b) answer distribution, and (c) annotation retrieval independence (closely tied with the notions of "open" and "closed" domain). TyDi provides a different set of natural questions per language, at the expense of direct comparability across languages. Not only are the TyDi questions different between languages, but the percentage of answerable passages varies dramatically, from 22% in Korean to 69% in Arabic. XorQA-TyDi (Asai et al., 2021) partially resolves this issue by sourcing answers from English documents, but this may in turn re-introduce cultural biases. This suggests that the conceptual difficulty of these questions may also vary dramatically, as consumers from different locales cater their questions based on their existing beliefs of the quality of the virtual assistants in their language. As a result, it is difficult to interpret the core reasons why multilingual system's performance varies between languages. To ensure this property, MKQA verifies its questions are predominantly geographically invariant, and thus the answers will not change due to geographical or cultural factors.

The second difference between datasets is the answer distribution. MKQA answers (a) are predominantly entities (42.2%) or atomic answers such as dates, binary, or numbers with units, and (b) use a different definition of "Unanswerable". Xor-QA focuses only on answerable queries, TyDi's definition conditions on the presence of the answer in the passage, whereas MKQA's definition is based on the ability of a human to find a succinct answer to a question on the web, i.e. whether it is human answerable. As a result, our annotations are not limited by the quality of selected passages, and provide higher answer coverage (67.58% as opposed to the TyDi language average of 38%).

Finally, while MKQA does not expect an answer to be derived from a single source document, TyDi is an extractive QA dataset. Consequently, its answer annotations are defined as spans, tied directly to particular Wikipedia documents and fixed index from which they were retrieved. As an evaluation set we contend the flexibility of document-independent answers is critical to not restrain what approaches can be evaluated in future research.

3 Dataset Collection

We aim for certain properties of our evaluation set: (i) realistic questions, (ii) reliable annotations (e.g. via inter-annotator agreement), and (iii) a flexible task setup that makes as few assumptions as possible about the underlying modeling techniques, enabling fair comparison between any approach.

3.1 Query Selection

Our evaluation set collection pipeline begins with the Answer Curation steps outlined in Figure 1. These are designed to yield high-concensus answer labels, with normalized textual formats, expressive alias sets for robust comparison, and grounding in structured information for entity disambiguation or more informative analysis. For the first step, we sample 10,000 queries from Natural Questions (NQ) (Kwiatkowski et al., 2019), as this is one of the few QA datasets based on realistic queries, generated by information seeking users.

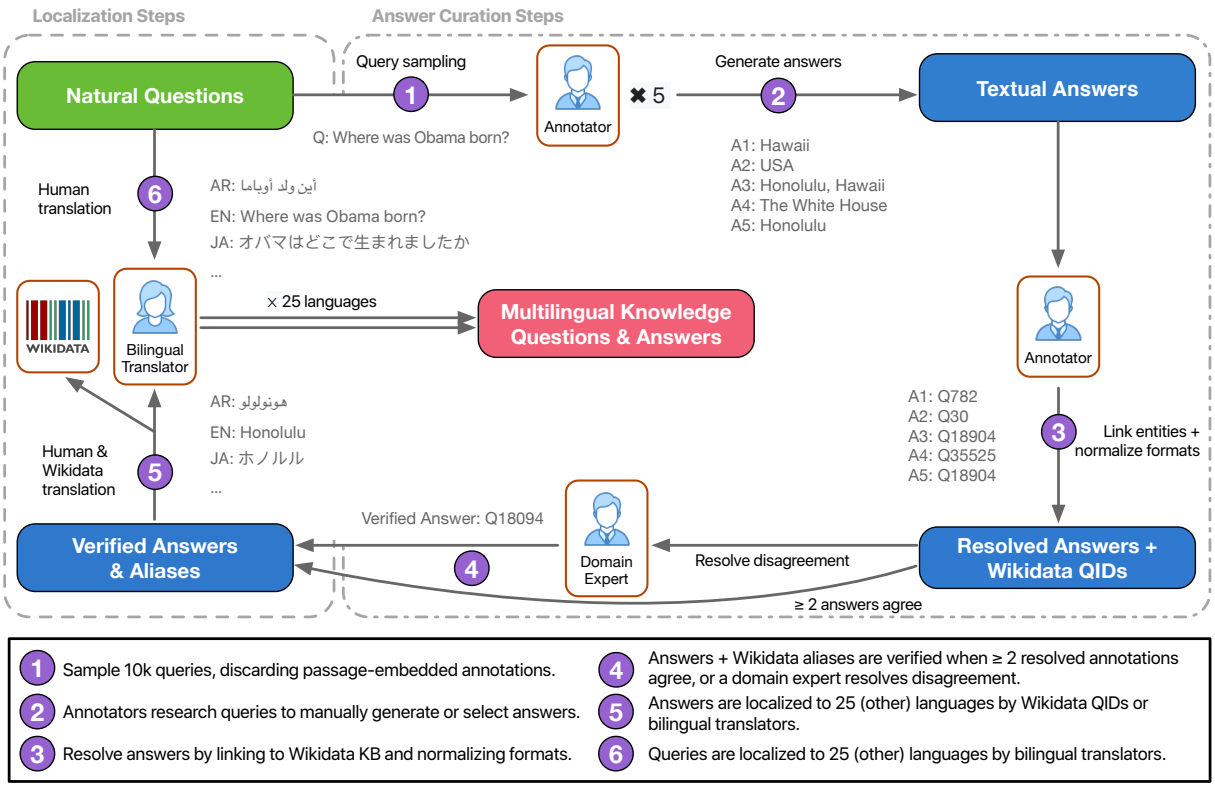

Figure 1: **Data Collection Process.** A depiction of the 6 sequential steps in our data collection pipeline. The first four steps involve Answer Curation, and the last two localize questions and answers into 26 target languages.

3.2 Raw Answer Collection

At the raw answer collection stage, 5 annotators are independently shown the query and asked to search the web to either copy *or generate* an ideal answer. They are asked to select an answer type (radio buttons) from the options shown below, and input the answer (text box) according to format instructions per answer type. The formatting constraints allow us to automatically link WikiData entities for the units in "number with units" and to gather well-structured data for answers such as dates, to save annotator time.

For each query, the graders select a typed answer from the following taxonomy:

- **Atomic value:** This category includes dates, numbers and number ranges with or without a unit (meters, years, ...).
- **Entities:** Entities are annotated with Wikidata QIDs and include generic entities, people, objects, and most locations.
- **Yes/No:** Type representing yes/no answers.
- **Short answer:** Answers which cannot be encapsulated in an atomic value, entity or binary (yes/no) answer, but are still a short phrase.
- **Long answer:** The long answer category indicates no simple factual answer or short phrase answers this question and a longer or visual explanation is required. During evaluation we treat these as "Unanswerable" for simplicity.
- **Unanswerable:** This category indicates that the query is not answerable, potentially because it is ill-formed or because no clear answer is available.

3.3 Answer Resolution

Given the query and a candidate answer from the previous stage, annotators are next asked to normalize date/number formats and resolve the answer text against Wikidata entities, where feasible. To resolve short textual answers against Wikidata entities, we apply an internal entity linking system to the answer string to generate Wikidata candidate entities.[3] The top 10 entity suggestions and their descriptions, along with the original query and short answer are then presented to 3 graders, who are asked to pick the correct reference entity or "None of the above." In cases where graders

[3]This step can be replicated using an off-the-shelf entity linker such as spaCy available at https://spacy.io/api/entitylinker.

do not achieve sufficient agreement or where the correct entity is not in the list, a domain expert (one of the MKQA authors/designers) provides the correct reference. Overall, this step enables us to disambiguate homonyms and collect valid answer synonyms/aliases, for more robustly measuring annotator agreement and prediction accuracy.

3.4 Answer Verification

Up until this stage, 5 raw answers were collected per query, and subsequently format normalized and resolved against Wikidata. In the fourth stage of Answer Curation (in Figure 1) any normalized answer given by at least 2 annotators is admitted to the final set as a gold answer. For those annotations that did not achieve the required agreement from at least two annotators, a domain expert (one of the MKQA authors/designers) with access to all 5 preliminary annotations is tasked to provide a final decision. This second manual round was afforded as much time per decision as necessary to obtain a satisfactory answer. The instructions permit the selection of existing normalized answer(s), modifying them slightly, or overriding them if necessary.

3.5 Answer Localization

In the last two stages of MKQA curation shown in Figure 1 we translate, or "localize" the English queries and answers into the target languages. Given the special care we took to avoid them in our methodology, and since we only localize short answers and queries (no context passages), we believe translation artifacts are likely to be minimal in MKQA.

Verified answers are localized into the target language by a combination of methods. For Wikidata-resolved answers, we leverage Wikidata's names and aliases for the target language. These names and aliases are transcribed in the native alphabet where appropriate, reflecting the expected answer in each language. Atomic answer types, including numeric, number with entity, and date types were also translated by this method, maintaining Arabic numerals for all languages, but naturalizing unit terms such as "November", "century", "b.c", "acres", and "light years". For date types specifically, for every combination of year, month, and day, we generate template answers in each language, accommodating both American and European date formats, as well as numeric and written out versions for months.

In cases where a Wikidata link could not be found, or where answers were not available for a given language code, professional bilingual human translators were used to provide the native equivalent. For this task, human translators are given access to the English query, the English answer, and where available the Wikidata link and Wikipedia page for the entity. We found localization quality improved when bilingual translators are shown several examples prior to grading, covering each of the localization options:

Localization Options:

- **Transliteration** is a type of conversion of a text from one script to another that involves swapping letters (thus trans- + liter-) in predictable ways (such as $\alpha \rightarrow$ a, $\chi \rightarrow$ ch, or æ $\rightarrow$ ae).
- **Translation** is the communication of the meaning of a source-language text by means of an equivalent target-language text.
- **Unchanged** is selected if the entity name does not need to be localized as it is commonly used as is.
- **Mix transliteration/translation/unchanged** if the entity is localized using more than one technique.

3.6 Query Localization

The final stage of MKQA construction, as shown in Figure 1, is query localization. As with answer localization, bilingual translators were asked to translate each query ensuring the query's meaning is maximally preserved, while naturally phrased. Translators were further instructed to use localized names of named entities if they exist in the target language and to transliterate names otherwise. Our translators, who are native speakers of the target language, are verified to live in the targeted region and are required to pass an entrance exam to verify a high level of fluency in English. Translators received a standard hourly wage varying with the target region and were not compensated per completed task, as is usual with alternative public services such as Mechanical Turk. On average, around 16 translators participated in the translation of the 10k source queries from English into each target language.

4 Dataset Quality and Analysis

Given our dataset collection and methodology, we evaluate the effect of our choices, and the properties